

Review Article

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Bladder Digital Twins for Neuromodulation: Current Platforms and a Roadmap to Closed-Loop Therapy

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Bladder disorders such as overactive bladder and neurogenic bladder impose a major symptom burden, yet neuromodulation therapies are largely delivered with fixed open-loop settings and limited adaptation to time-varying bladder states. Bladder digital twins are patient-anchored computational models linked to sensing streams and updated via state estimation and/or data assimilation. They offer a pathway to closed-loop personalized therapy, but bladder implementations remain fragmented across modeling, sensing, and stimulation. This review synthesizes digital-twin technologies relevant to bladder physiology, sensing, and neuromodulation and translates lessons from mature organ digital twins to bladder requirements for closed-loop control. We conducted targeted searches of PubMed, Web of Science, IEEE Xplore, and Google Scholar (2013–2025). We reviewed digital-twin paradigms for the heart, brain, lungs, liver, and kidney and summarized bladder approaches including finite-element bladder-wall mechanics, conductivity-based torso models for wearable bioimpedance optimization, strain-to-geometry reconstruction using stretchable sensors, and sensor-informed closed-loop control. Clinically deployable twins pair mechanistic structure with limited reliable data streams and use hybrid estimation for indirect, noisy measurements. In bladder applications, feasibility has been demonstrated for state estimation and model-guided design; key barriers include ambulatory motion artifacts and impedance drift, sparse clinical anchors for personalization, limited longitudinal human validation, and incomplete safety engineering for stimulation control. A roadmap toward closed-loop bladder digital twins should prioritize ambulatory-grade sensing with artifact handling and recalibration, uncertainty-aware real-time state estimation, model-informed stimulation optimization, and fail-safe control with standardized reporting of accuracy, latency, uptime, and drift.

Keywords: Urinary bladder; Electrical stimulation; Patient-specific modeling; Computer simulation; Feedback, Physiologic; Monitoring, Physiologic

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INTRODUCTION

Digital-Twin Technology in Healthcare and Precision Medicine

A healthcare digital twin is a patient-specific virtual model that

integrates multimodal data including clinical records, molecular profiles, medical imaging, genetic information, and environmental factors to simulate, monitor, and predict health states and treatment responses. A digital twin comprises the physical patient, a computational representation, and a bidirectional

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data connection (continuous updating from real-world measurements). By combining sensor and Internet of Things inputs (wearables/implants and connected devices) with multiscale computational models, digital twins can forecast disease trajectories and treatment outcomes [1, 2]. This supports precision medicine by enabling *in silico* testing before *in vivo* use and detecting early deviations to trigger timely intervention [3].

In this review, we distinguish three related terms. A computational model is any mechanistic or data-driven framework that simulates anatomy or function but is not necessarily individualized or updated over time. A patient-specific model is calibrated to an individual using subject data (e.g., imaging-derived anatomy or urodynamic parameters) but may remain static after calibration. We use the term digital twin more strictly to denote a patient-anchored model that is linked to measurement streams and includes an explicit mechanism for state estimation and/or data assimilation (updating the model using new data) to update latent state and/or parameters over time, enabling tracking and prediction of time-varying physiology. Accordingly, we refer to many bladder studies as digital-twin components unless they include repeated or continuous measurements together with an explicit estimation or update loop.

Rationale for This Review

Digital-twin approaches have advanced across multiple organ systems by integrating patient-specific imaging, sensor data, and computational modeling to support diagnosis, risk stratification, and intervention planning in cardiology, neurology, pulmonology, and related fields [2]. In contrast, bladder digital-twin applications remain underdeveloped despite the prevalence and burden of overactive and neurogenic bladder disorders [4]. Neuromodulation therapies, including percutaneous tibial nerve stimulation and sacral nerve stimulation, are typically delivered using fixed open-loop regimens with limited adaptation to time-varying physiological states [4, 5]. Recent studies demonstrate feasibility for key enabling components of bladder digital twins, including real-time bladder volume sensing, finite-element (FE) modeling of detrusor mechanics, and early sensor-driven control strategies [6–8]. These efforts remain fragmented across modeling, sensing, and stimulation, limiting translation to closed-loop therapy. This review consolidates digital-twin methodologies from major organ systems and applies them to bladder neuromodulation, summarizing current advances, identifying barriers, and proposing a roadmap toward safe clinically deployable closed-loop bladder therapy.

Scope and Nature of the Review

This narrative review summarizes digital-twin technologies relevant to bladder physiology, sensing, and neuromodulation. We performed targeted searches in PubMed, Web of Science, IEEE Xplore, and Google Scholar, prioritizing peer-reviewed journal articles (2013–2025). Studies were chosen to highlight key technical and translational advances in organ-level frameworks, bladder modeling/sensing, and closed-loop neuromodulation. Because this is a narrative review, study selection was not preregistered or systematic; when evidence overlapped, we prioritized relevance and recency.

DIGITAL TWINS OF MAJOR HUMAN ORGANS

This section summarizes representative digital twins for the heart, brain, lungs, liver, and kidney, focusing on modeling frameworks, primary data inputs, and translational readiness. Detailed technical descriptions are consolidated in the following section and Table 1.

Heart

Cardiac digital twins support procedural planning (e.g., transcatheter valves, catheter ablation) and arrhythmia risk assessment [9, 10]. The Living Heart Project (Fig. 1A–C) established whole-organ electromechanics using computed tomography/magnetic resonance imaging (CT/MRI)-derived anatomy and FE modeling (physics-based tissue simulation) [9], and more recent implementations demonstrate GPU (graphics processing unit)-accelerated (hardware to speed computation) electromechanics and flow simulation at clinically relevant timescales (Fig. 1D–I) [11]. Personalization is commonly anchored by imaging with limited physiological measurements, including late gadolinium enhancement MRI for fibrosis-informed substrate modeling and inverse methods (estimating physiology from outputs) calibrated to pressures when available [10, 12]. These workflows show that mechanistic models can be individualized with sparse anchors, relevant to bladder systems with indirect sensing.

Brain

Brain digital twins model large-scale network dynamics for clinical research applications such as epilepsy and movement disorders (Fig. 2A) [13, 14]. Representative frameworks link diffusion MRI-derived connectomes (brain wiring maps) to neural-mass or spiking models (simplified population/neuronal

Table 1. Cross-organ comparison of digital-twin frameworks

Organ	Core model type	Primary data inputs	Maturity level	Representative platform
Heart	Electromechanical finite-element modeling; fluid-structure CFD	CT/MRI geometry (including late gadolinium enhancement MRI when available), ECG traces, pressure waveforms (inverse calibration)	Clinical pilots (in-hospital)	Living Heart Project [9]; GPU-based heart twin [11]
Brain	Neural-mass and network models	DTI connectome, EEG/MEG/ECOG, fMRI	Early clinical research	The Virtual Brain (Charité) [16]
Lung	Electrical impedance tomography-guided finite-element modeling; CFD; sensor-fusion machine learning	CT scans, electrical impedance tomography, spirometry, peripheral oxygen saturation (SpO ₂), environmental sensors	Proof-of-concept studies	Lung digital-twin framework [18]; EIT twin-guided reconstruction [19]; DT-enabled IoMT system [20]; structured-report twin pipeline [21]
Liver	Agent-based regeneration; 0D (lumped-parameter) circulation; ODE-network models	CT/MRI volumes, hemodynamics, laboratory biochemistry	Preclinical and pilot trials	Agent-based lobule model [22]; hepatectomy risk twin [23]; virtual liver metabolism twin [24]; AR-guided puncture twin [25]; personalized nutrition twin [26]
Kidney	3D anatomical meshes; metabolic flux networks; imaging classifiers	Contrast CT, routine laboratories, blood pressure, radiology reports	Early validation (small cohorts)	3D anatomical navigation twin [27]; GMF-DT CKD predictor [28]

This table summarizes representative digital twin approaches across 5 organ systems, focusing on dominant model classes, primary patient data inputs, typical maturity level, and example platforms.

CFD, computational fluid dynamics; CT, computed tomography; MRI, magnetic resonance imaging; ECG, electrocardiogram; GPU, graphics processing unit; DTI, diffusion tensor imaging; EEG, electroencephalography; MEG, magnetoencephalography; ECOG, electrocorticography; fMRI, functional magnetic resonance imaging; EIT, electrical impedance tomography; GMF-DT, generalized metabolic flux digital twin; IoMT, Internet of Medical Things; 0D, lumped-parameter model; ODE, ordinary differential equation; AR, augmented reality; 3D, 3-dimensional; CKD, chronic kidney disease.

dynamics) (Fig. 2B) and use multimodal measurements for personalization and validation (Fig. 2C) [15-17]. The Virtual Brain has reported clinical use in presurgical contexts (Fig. 2A) [16]. These systems highlight uncertainty-aware estimation under indirect and noisy measurements, a central requirement for bladder-state inference.

Lungs

Lung digital twins emphasize continuous monitoring and simulation under time-varying clinical and environmental conditions (Fig. 3A) [18]. Monitoring twins fuse wearable and ambient sensing for real-time tracking (Fig. 3B) [18], while simulation twins use imaging-derived multiphysics models that can improve reconstruction robustness in impedance-based monitoring (Fig. 3C) [18, 19]. Personalization combines CT-defined anatomy with trends in impedance and SpO₂, sometimes incorporating longitudinal clinical data [18-21]. These approaches motivate bladder pipelines that prioritize artifact handling (motion/posture effects, drift, missing data), drift control, and reliability reporting under real-world variability.

Liver

Liver digital twins integrate regeneration, biomechanics, and circulation modeling to support surgical planning and outcome prediction after partial hepatectomy [22, 23]. Representative examples include agent-based regeneration modeling (Fig. 3D) [22] and circulation models that forecast portal pressures after resection [23]. Additional paradigms include ordinary differential equation (ODE)-network models for metabolic and signaling pathways and procedure-oriented targeting under motion during interventions [24-26]. These studies motivate layered bladder twins that connect longer-term remodeling or compliance changes to functional outcomes.

Kidney

Kidney digital twins often combine patient-specific anatomy with predictive inference to support surgical navigation and earlier risk forecasting [27, 28]. Anatomical twins reconstruct renal structures from contrast CT into navigable 3-dimensional models for robotic partial nephrectomy (Fig. 3E and F) [27]. Predictive twins integrate routine biomarkers and vital signs for

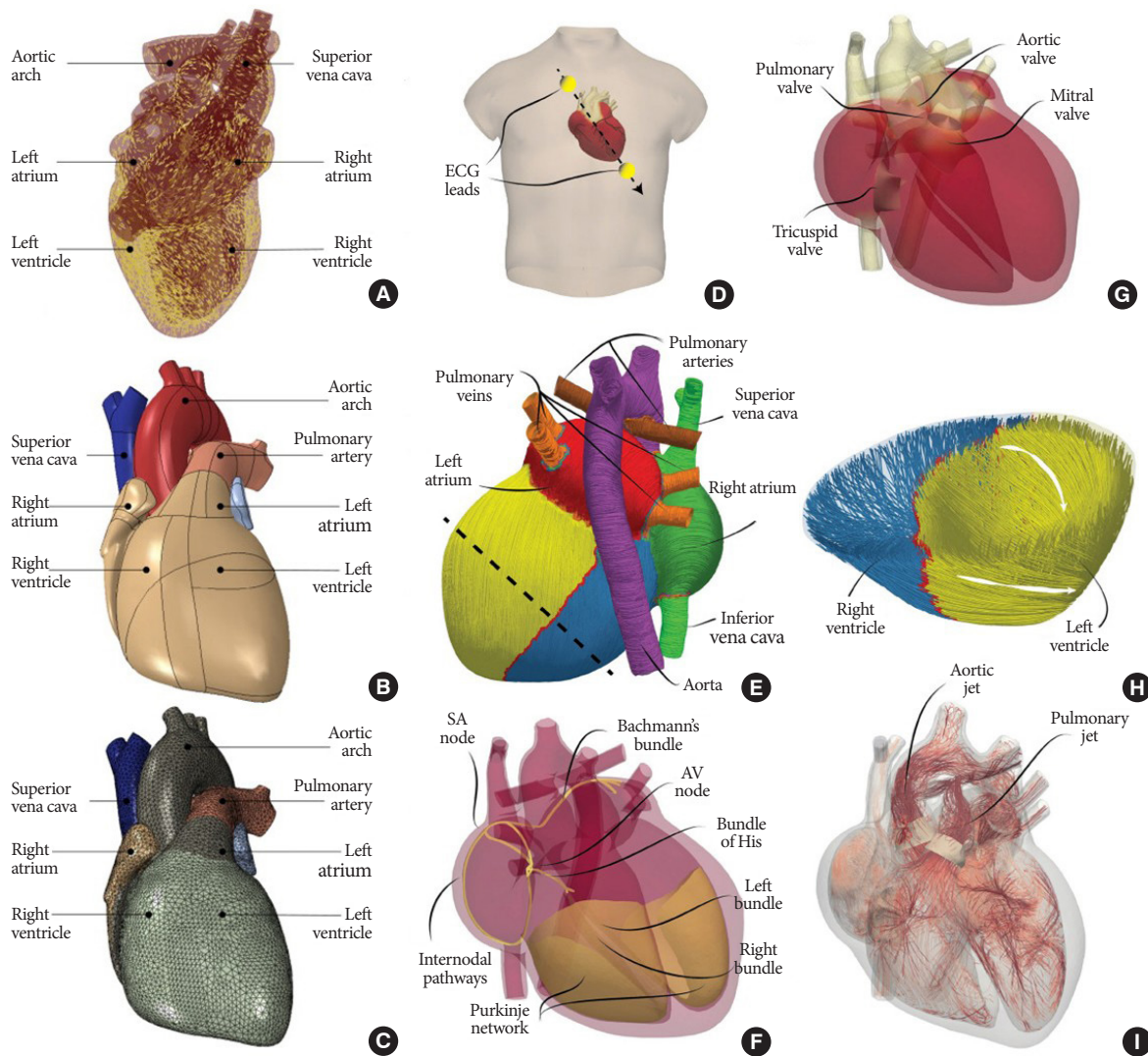


Fig. 1. Digital twin of the human heart. (A–C) Living Heart Project [9]: (A) Anatomical 3-dimensional heart model showing the aortic arch, pulmonary artery, superior vena cava, and all 4 chambers. (B) Finite-element mesh comprising 208,561 tetrahedral elements, 47,323 nodes, and 189,292 total degrees of freedom. (C) Myocardial fiber orientation map with 208,561 discrete fiber and sheet directions assigned to each integration point. Reprinted from Baillargeon et al. *Eur J Mech A Solids* 2014;48:38-47 [9], under the Creative Commons CC BY-NC license. (D–I) GPU (graphics processing unit)-based heart model [11]: (D) Heart positioned in torso with virtual electrocardiogram (ECG) leads. (E) Anatomical assembly showing vasculature and chambers. (F) Electrophysiological conduction system including sinoatrial (SA) node, atrioventricular (AV) node, and Purkinje network. (G) Valve anatomy of the 4 cardiac valves. (H) Ventricular wall fiber orientation cross-section. (I) Simulated systolic flow streamlines including aortic and pulmonary jets. Reprinted from Viola et al. *Sci Rep* 2023;13:8230 [11], under the Creative Commons CC BY 4.0 license.

longitudinal chronic kidney disease prediction, and imaging-based classifiers are also explored in research settings [28, 29]. These workflows reinforce a practical template for bladder translation, coupling anatomy with routinely available anchors when continuous high-fidelity sensing is limited.

CROSS-ORGAN COMPARISON AND KEY METHODOLOGIES

Comparative Summary

Table 1 compares digital twins across 5 organs by model type, key data inputs, maturity, and representative platforms. More

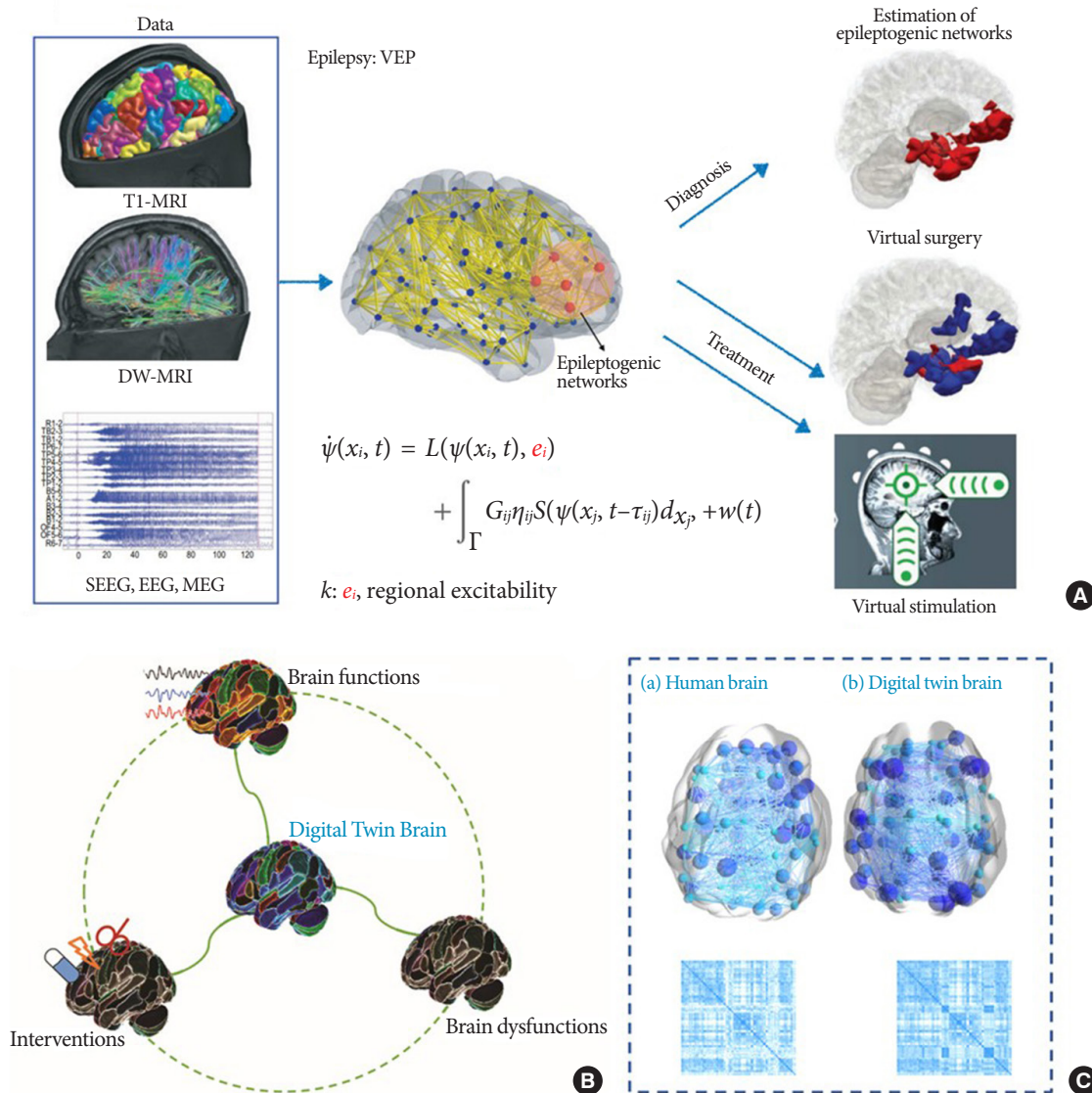


Fig. 2. Brain digital twin applications. (A) Workflow of the Virtual Epileptic Patient from The Virtual Brain [16], showing patient-specific whole-brain network modeling, connectome-driven dynamics, and parameter estimation for seizure forecasting. Reprinted from Wang et al. Natl Sci Rev 2024;11:nwae079 [16], under the Creative Commons CC BY-NC license. (B) Schematic of the Digital Twin Brain framework [15], showing integration of structural and functional data to model normal cognition, neurological pathology, and therapeutic interventions including deep brain stimulation, transcranial magnetic stimulation, and pharmacological agents. Reprinted from Xiong et al. Intell Comput 2023;2:0055 [15], under the Creative Commons CC BY 4.0 license. (C) Topology comparison in DTBVis [17], showing network graphs and functional connectivity matrices contrasting the human connectome with digital twin outputs for validation against functional magnetic resonance imaging/diffusion tensor imaging data. Reprinted from Li et al. Vis Inform 2023;7:41-53 [17], under the Creative Commons CC BY-NC license. VEP, virtual epileptic patient; T1-MRI, T1-weighted magnetic resonance imaging; DW, diffusion-weighted; SEEG, stereoelectroencephalographic; EEG, electroencephalography; MEG, magnetoencephalography.

clinically mature systems typically combine physics-based models with few reliable routine measurements, whereas earlier platforms favor data-driven inference. This trend is relevant to

bladder twins, where sensing is sparse and personalization should use mechanistic models plus limited clinical references.

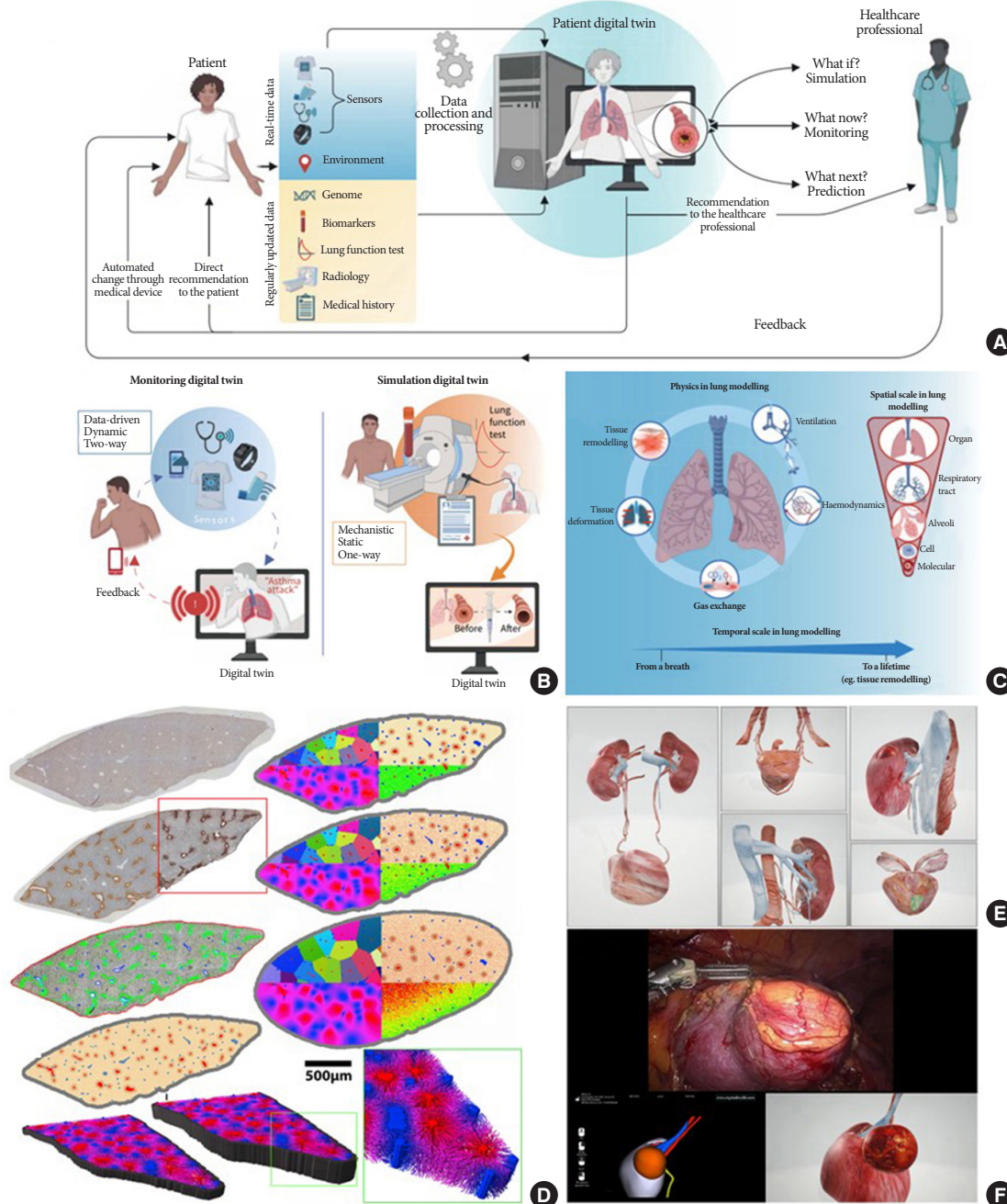


Fig. 3. Digital twins for chronic lung diseases and liver regeneration modeling and kidney surgical navigation. (A) Components of a patient digital twin system for respiratory diseases. (B) Comparison of monitoring and simulation digital twins. (C) Multiphysics and multiscale aspects of lung modeling for simulation digital twins. Reprinted from Gonsard et al. *Eur Respir Rev* 2024;33:240159 [18], under the Creative Commons CC BY 4.0 license. (D) Agent-based liver digital twin simulating biomechanical regeneration after partial hepatectomy [22], showing construction of a computational single-cell-based model from whole-slide scans using an image processing and analysis pipeline. Reprinted from Hoehme et al. *iScience* 2022;26:105714 [22], under the Creative Commons CC BY-NC license. (E and F) Anatomical digital twin for intraoperative navigation during robotic partial nephrectomy [27]: (E) Final rendering of 3-dimensional (3D) virtual digital twin models of urological organs with realistic textures captured intraoperatively. (F) Intraoperative comparison showing a texture-less 3D model versus a realistic digital twin model of the same organ. Reprinted from Amparore et al. *BJUI Compass* 2025;6:e453 [27], under the Creative Commons CC BY-NC license.

Core Modeling Techniques

Across organs, digital twins rely on a recurring toolset. FE and computational fluid dynamics models support physics-based mechanics and flow simulation for deformation, transport, and fluid-structure interactions [9, 11, 18, 19]. Lumped-parameter (0D) models (fast simplified dynamics) provide computationally light system dynamics suitable for fast prediction and control-oriented representations [23, 28]. Electrophysiology and network models, including neural-mass and connectome-driven frameworks, capture activation and large-scale brain dynamics [15, 16]. Agent-based and ODE-network models represent remodeling, regeneration, and pathway-level mechanisms [22–24]. Machine learning is widely used for prediction from imaging and sensor streams and as a component in hybrid pipelines [18, 19, 28, 29]. Increasingly, hybrid twins combine mechanistic models with learned surrogates (fast machine learning approximations of simulations) and data assimilation to enable personalization, uncertainty handling (confidence-aware estimation), and near real-time updates [12, 16, 18, 28]. Clinically oriented personalization often relies on imaging-defined anatomy with sparse physiological anchors, e.g., fibrosis-informed cardiac substrates from late gadolinium enhancement MRI and inverse calibration to pressure or waveform constraints when available [10, 12]. These strategies provide a template for bladder twins that must operate with limited clinical reference measurements and indirect sensing.

For bladder neuromodulation, these approaches match implementation needs. FE models capture bladder-wall mechanics, compliance, and pressure-volume behavior, while conductivity and bioimpedance models support volume estimation and electrode optimization [6]. 0D bladder models provide computationally light state evolution suitable for embedded control, and learned surrogates reduce FE-level computation to meet latency constraints. Because bladder state is typically inferred from indirect signals, hybrid estimation is often the most practical pathway to robust personalization across patients and daily-life conditions.

Sensor and Data Integration Challenges

Across organ systems, model-guided reconstruction and sensor fusion improve robustness under noise and variability, reinforcing the importance of artifact suppression, drift management, and missing-data tolerance for ambulatory bladder monitoring [18–20]. Digital twins must align heterogeneous data streams in time, suppress noise, manage missing data, and integrate imag-

ing pipelines that require segmentation and mesh generation. Wearable and implantable sensors add drift (slow baseline change over time), sampling variability, and device-specific artifacts (motion/contact-related noise), while secure storage and transfer must meet privacy requirements.

Bladder digital twins are especially sensitive to ambulatory confounds. Posture shifts, motion artifacts, electrode displacement, skin–electrode impedance changes, and long-term drift can degrade volume or pressure estimates and increase false stimulation triggers [6, 8]. Bladder-oriented pipelines should therefore include artifact suppression (removing motion-related noise), gap handling (missing-data robustness), posture and activity normalization (adjusting for body position/movement), and periodic recalibration (anchoring to clinical reference measurements) such as urodynamics or ultrasound. For reproducibility, studies should report at minimum estimation error, end-to-end latency (time from sensing to stimulation), uptime, and drift over time under real-world use.

Common Gaps and Future Needs

Across organ systems, many digital twins lack longitudinal human validation, and true real-time bidirectional coupling remains uncommon outside controlled settings. Scalability is limited by inconsistent data schemas and scarce benchmarking, while regulatory expectations for software-based medical devices continue to evolve.

For closed-loop bladder neuromodulation, three gaps are central. First, prospective ambulatory human validation is needed with endpoints that capture both physiological accuracy and symptom benefit in daily life. Second, safety engineering must be explicit, including stimulation limits, uncertainty-aware triggering, failure detection (dropout, drift, model mismatch), and safe fallback behavior suitable for neuromodulation [8]. Third, standardized benchmarks, reporting templates, and interoperable schemas spanning sensing, imaging, and urodynamics are required to compare platforms and support translation.

BLADDER DIGITAL TWINS AND NEUROMODULATION

This section summarizes bladder physiology and clinical motivation, then reviews modeling approaches, sensing technologies, neuromodulation integration, personalization strategies, and representative bladder digital-twin platforms.

Bladder Physiology and Clinical Drivers

The urinary bladder stores and expels urine through coordinated interactions between the detrusor smooth muscle, urothelium, urethral sphincter, and central and peripheral neural circuits. During filling, urothelial stretch activates afferent A δ fibers that relay bladder-wall tension to spinal and supraspinal centers, while inhibitory pathways suppress detrusor contraction and maintain continence. Voiding occurs when cortical inhibition is released, triggering a reflex detrusor contraction with synchronized sphincter relaxation [30].

Disruption of these control loops underlies common bladder disorders. Overactive bladder is characterized by urgency, frequency, nocturia, and involuntary detrusor contractions at low volumes, often driven by heightened afferent excitability, altered smooth muscle activity, or urothelial mediator release such as adenosine triphosphate. Stress and urge incontinence result from pelvic support failure or impaired sphincter function, whereas neurogenic bladder arises from neurological conditions that disrupt communication between the bladder and central nervous system [31]. For closed-loop therapy, these mechanisms translate into measurable control variables, including bladder volume, pressure trend, risk of detrusor overactivity, and sphincter activity that can inform stimulation timing and dosing.

Bladder Digital-Twin Modeling Approaches

Current bladder digital twins span complementary computational strategies. The bladder outlet obstruction mechanobiology (BOOM) framework, reported in a COMSOL (commercial multiphysics simulation software) technical report/paper entry, models bladder-wall mechanics using a multilayer FE formulation with constrained-mixture growth and remodeling, integrating passive matrix, collagen fiber, and smooth muscle stresses, and calibrating parameters using urodynamic pressure-volume loops and histological composition [32].

Crane et al. [6] developed a conductivity-based bladder twin that segments a patient-specific torso mesh and assigns frequency-dependent tissue conductivities using Cole–Cole models (frequency-dependent tissue electrical properties); by morphing bladder geometry (changing shape across filling states), the model generates sensitivity maps (where measurements are most informative) to optimize bioimpedance electrode configurations (Fig. 4A–G). Cho et al. [7] introduced a strain-to-geometry (deformation-to-shape reconstruction) twin in which stretchable gold-carbon nanotube sensors map anisotropic wall expansion to reconstructed bladder geometry for volume esti-

mation and feedback applications; this report is currently available as a preprint. Egorov's digital twin reconstructs pelvic anatomy from MRI and uses machine-learning components to estimate tissue stiffness and activation patterns under loading, supporting functional assessment of pelvic support [33]. Collectively, these platforms can be positioned as follows: BOOM emphasizes mechanics and remodeling, Crane focuses on measurement optimization for volume inference, Cho targets strain-based state reconstruction, and Egorov models pelvic support behavior relevant to continence.

Sensor Technologies for Bladder Digital Twins

Sensing is central to bladder digital twin operation and involves tradeoffs among invasiveness, signal fidelity, and long-term stability. Wearable bioimpedance systems (small current/voltage measurements across electrodes) estimate bladder volume by injecting low-amplitude alternating currents and measuring voltage changes across electrode pairs; Crane et al. [6] reported mean absolute error of 6.1 mL for volumes below 100 mL and approximately 1 percent error above 100 mL under their evaluation conditions. Stretchable AuCNT strain sensors (gold-carbon nanotube composites) mounted on compliant substrates measure anisotropic wall deformation and can support state reconstruction, but their long-term stability and chronic-use robustness remain key translational questions [7]. Implantable pressure sensors can provide direct pressure tracking with high fidelity and may support long-term contraction monitoring, while less-invasive suprapubic microelectromechanical systems sensors aim to reduce catheter dependence; however, chronic drift, biofouling (protein/cell buildup), and encapsulation (tissue response around the device) effects must be managed [8]. Additional modalities include sphincter electromyography as a proxy for outlet function, dorsal root ganglion recordings for decoding in animal models, and near-infrared spectroscopy for detrusor hemodynamic monitoring during filling [8]. Bladder digital twins need multimodal sensing to balance reliability and feasibility. Overall, wearables favor safety and scalability but face motion/impedance drift, whereas implantables offer higher fidelity but require long-term packaging and stability solutions [6–8].

Neuromodulation Integration and Closed-Loop Control

Integrating neuromodulation with bladder digital twins enables adaptive therapy by linking continuous sensing to model-based state estimation and stimulation control. In principle, the twin

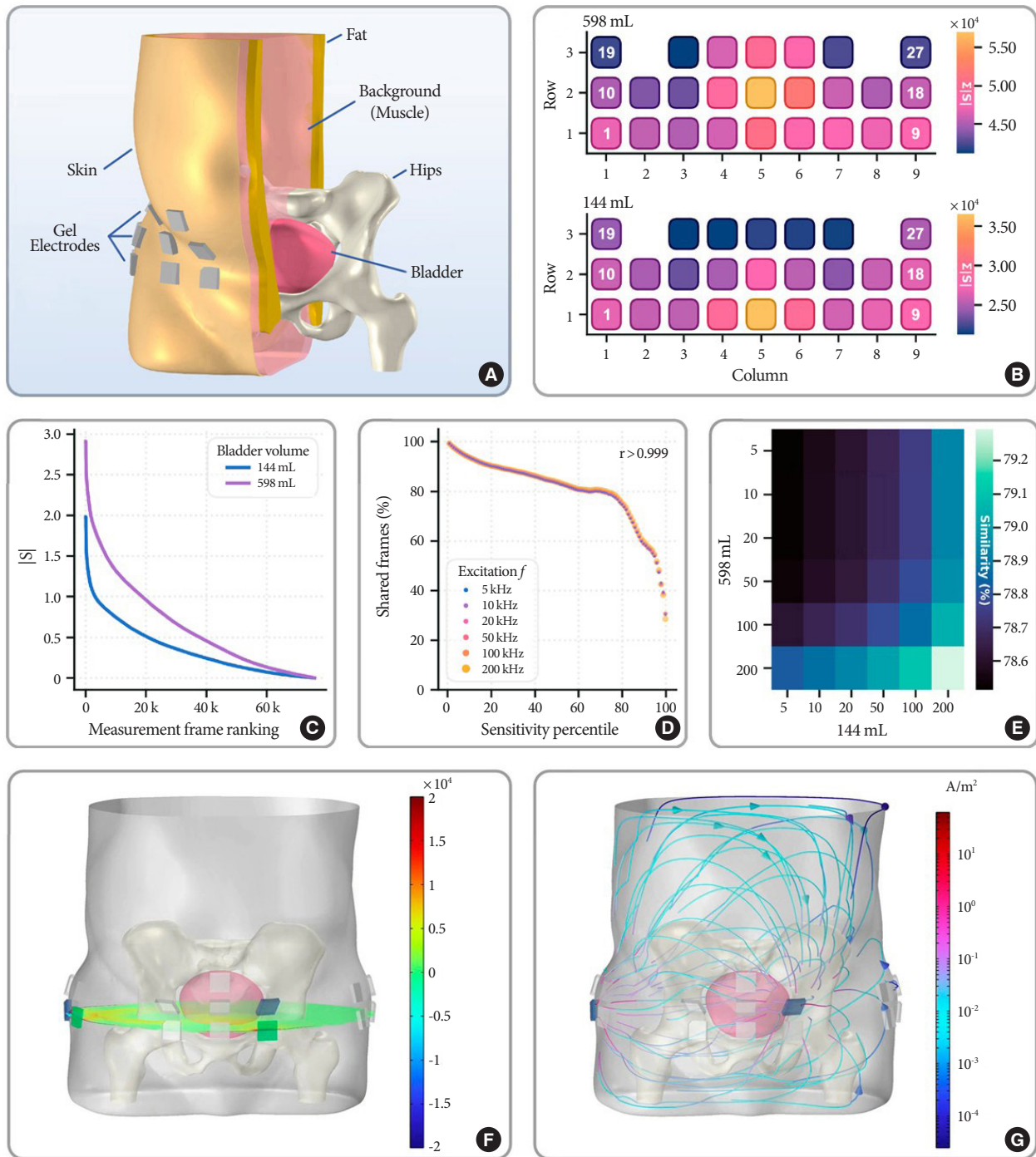


Fig. 4. Sensitivity analysis of a bladder digital twin for optimized electrode configurations and volume estimation [6]. (A) Segmented anatomical and modeling domains from patient computed tomography/magnetic resonance imaging data. (B) Electrode-pair sensitivity matrix computed across all configurations. (C) Spatial distribution of absolute sensitivity in the pelvic cavity. (D) Frame similarity heatmap across bladder volumes and stimulation frequencies. (E) Shared frame matrix identifying volume-independent electrode placements by frequency. (F) Three-dimensional view of the digital twin highlighting the sensitivity distribution through the internal tissues. (G) Three-dimensional view of the digital twin visualizing current density at 50 kHz. Reprinted from Crane et al. NPJ Digit Med 2025;8:73 [6], under the Creative Commons CC BY 4.0 license.

can predict rising urgency or imminent detrusor overactivity and adjust stimulation in real time, moving from fixed open-loop regimens toward on-demand intervention [8, 34].

Current clinical neuromodulation remains largely open-loop (fixed settings without feedback). Percutaneous tibial nerve stimulation delivers fixed pulse trains during scheduled sessions [35], with symptom improvements commonly reported without physiological feedback; implantable tibial stimulators similarly operate without continuous physiological feedback [4, 36]. Sacral nerve stimulation is typically tuned empirically and adjusted during clinic visits, while interferential current therapy generally uses predefined frequencies and electrode placements without patient-specific field optimization [4, 5, 37].

For a closed-loop bladder digital twin, the core control variables include stimulation trigger timing, intensity, duration, and waveform or pattern selection, depending on modality and device capabilities. Candidate feedback signals include pressure trend or contraction markers, bioimpedance-derived volume trajectories, and sphincter or pelvic-floor activity proxies such as electromyography when available [6, 8]. Because detrusor events and urgency escalation occur on clinically meaningful timescales, the end-to-end latency from sensing to state estimation to stimulation delivery should be specified and validated so that control actions occur early enough to prevent symptoms while avoiding excessive or unnecessary stimulation. A model-informed twin can additionally support *in silico* parameter screening, selecting electrode configurations and stimulation settings that maximize efficacy while preserving comfort and safety constraints [8, 34].

Personalization Strategies

Personalization in bladder digital twins relies on patient-specific calibration and periodic updating. BOOM adapts remodeling parameters using individual urodynamic pressure-volume loops and histological measurements [32]. Crane's workflow personalizes anatomy and tissue conductivities from imaging to optimize electrode configurations for bioimpedance sensing [6]. Cho calibrates strain-to-geometry mappings based on subject-specific volume measurements, although long-term recalibration under chronic use remains an open issue [7]. Egorov's supervised learning adjusts stiffness and activation parameters using imaging and patient outcomes [33]. Across platforms, a practical approach is to combine continuous sensor-driven adaptation with scheduled recalibration checks to detect drift and to refresh clinical anchors (e.g., periodic urodynamics or ultra-

sound snapshots) when sensor behavior or clinical status changes.

Representative Bladder Digital-Twin Platforms

Table 2 summarizes current bladder digital-twin platforms, including model type, sensing modality, mechanism, reported outcomes, and remaining challenges. Together, these studies demonstrate feasibility for modeling bladder mechanics and estimating bladder state, but also highlight that closed-loop readiness remains limited by gaps in ambulatory validation, long-term sensor stability, and explicit safety engineering for neuromodulation [6-8, 32, 33]. These limitations motivate the roadmap presented in the following section.

ROADMAP TO CLOSED-LOOP BLADDER DIGITAL TWINS

Translating bladder digital twins into clinically reliable closed-loop neuromodulation requires coordinated progress in sensing robustness, real-time state estimation, model-informed control, safety engineering, and prospective validation.

Ambulatory-Grade Sensing and Signal Robustness

The first priority is ambulatory sensing that remains stable under motion, posture transitions, hydration variability, and long-term drift [6, 8]. Wearable and implantable signals need standardized preprocessing for artifact suppression, gap handling, and recalibration strategies anchored to sparse clinical references such as urodynamics, ultrasound snapshots, or diary-linked voiding events [6, 8]. To enable fair comparison across platforms and define minimum requirements for closed-loop use, studies should consistently report estimation error, end-to-end latency, uptime (signal availability), and drift over time under real-world conditions [1, 3, 8].

Real-Time State Estimation and Data Assimilation

The second priority is state estimation (inferring bladder state from indirect signals) of clinically meaningful latent states, including bladder volume, pressure trends, and risk of imminent detrusor overactivity. Because measurements are often indirect and noisy, purely data-driven models may overfit across patients and contexts, while purely mechanistic models may be costly or unidentifiable [1-3, 12, 16]. Hybrid frameworks that couple physics-based dynamics with learning-based surrogates and data assimilation are ideal for practical systems, reflecting

Table 2. Summary of current digital-twin research for bladder

Attribute	Boniface et al. (2024) [32]	Crane et al. (2025) [6]	Cho et al. (2025) [7]	Egorov (2024) [33]
Evidence setting	Simulation (calibrated using clinical urodynamic anchors)	Human (controlled evaluation; anatomy from patient imaging as reported)	<i>Ex vivo</i>	Human (pilot cases)
Publication status	Technical report/web paper (COMSOL)	Peer-reviewed journal article	Preprint (SSRN)	Peer-reviewed journal article
Model type	Constrained-mixture finite-element (growth/remodeling)	AC/DC conductivity finite-element torso–bladder twin	Strain-to-geometry reconstruction twin	ML-augmented pelvic support finite-element
Sensor/inputs	Urodynamic pressure-volume loops; histology	Wearable bioimpedance electrode array	Stretchable AuCNT strain sensor array	Surface tactile-imaging pressure arrays
Mechanism	Fits growth-remodeling parameters from pressure-volume behavior and tissue composition	Optimizes electrode configurations and estimates volume from impedance-sensitive voltage patterns	Maps multichannel strain to reconstructed bladder geometry for volume estimation and feedback	Estimates tissue stiffness/activation related to pelvic support from pressure patterns
Reported performance (value)	Reproduced clinical pressure-volume loops and wall-stress patterns	Mean absolute error 6.1 mL (< 100 mL); ~1% error (≥ 100 mL)	<i>Ex vivo</i> volume error $\leq 11.6\%$; closed-loop latency < 10 ms	Matched clinical measurement within 1 mm in 85% of pilot cases
Performance context (endpoint)	Endpoint: agreement to urodynamic pressure-volume behavior and derived stress fields	Endpoint: bladder volume estimation error from bioimpedance	Endpoint: reconstructed volume error; closed-loop timing (compute/control latency)	Endpoint: agreement to clinical measurement/assessment metric
Performance context (assumptions/notes)	Model-fit performance depends on availability/quality of urodynamic and tissue composition anchors	Performance reported under evaluation conditions described in the study (electrode placement, posture/activity constraints, and calibration procedure)	<i>Ex vivo</i> setup; <i>in vivo</i> long-term stability and drift not yet established	Pilot cohort; performance depends on sensor placement and clinical reference standard used in the study
Next steps	Couple growth and voiding simulators; model progressive obstruction and therapeutic interventions	<i>In vivo</i> ambulatory trials with posture and hydration normalization	Validate chronic implant biocompatibility; implement long-term drift correction	Automate segmentation and mesh generation; extend to integrated bladder–floor twin

This table compares bladder-related digital-twin studies by evidence setting, publication status, model type, sensing modality, mechanism, reported outcomes, and stated next steps toward translation. Reported performance metrics are listed with context (setting, endpoint, and assumptions) as described in each source.

COMSOL, commercial multiphysics simulation software; SSRN, Social Science Research Network (preprint server); AC/DC, alternating current/direct current; ML, machine learning; AuCNT, gold-carbon nanotube.

how clinically oriented organ twins combine mechanistic structure with limited routine data streams [1, 2, 12, 16, 28]. Estimators should quantify uncertainty so that control actions remain conservative when confidence is low and can degrade safely when sensing quality changes [1, 3, 8].

Model-Informed Personalization and Stimulation Optimization

The third priority is surpassing heuristic open-loop therapy by incorporating patient-specific calibration and optimization of

stimulation settings [5, 35, 36]. Bladder digital twins should support personalization using imaging and clinical anchors and include simplified but testable mappings from stimulation parameters and electrode configurations to expected physiological effects, leveraging model-guided optimization approaches already demonstrated for bladder monitoring design and early closed-loop stimulation [6, 34]. Optimization should balance symptom reduction with stimulation burden, comfort, battery constraints, and safety margins, and should be evaluated using clinically relevant endpoints such as urgency episode reduction,

reduced stimulation time, and improved quality-of-life measures [5, 34-37].

Closed-Loop Control Design and Safety Engineering

The fourth priority is closed-loop control with explicit safety constraints and fail-safe behavior (restore safe default settings). Practical systems must anticipate sensor dropout, drift, model mismatch, and false triggers that can lead to inappropriate stimulation, which is critical given the known challenges in bladder biomarker sensing and the early-stage nature of closed-loop demonstrations [8, 34]. Closed-loop implementations should enforce limits on stimulation intensity and duration, implement refractory periods, detect sensor and model failures, and restore clinician-defined parameters when reliability is compromised [5, 35, 36]. Latency budgets from sensing to estimation to stimulation delivery should be defined and validated against the timescales of clinically meaningful bladder events to ensure control actions arrive early enough to prevent symptoms [34].

Fig. 5 illustrates a reference closed-loop workflow in which sensing streams are first stabilized by preprocessing and artifact handling, then fused through state estimation (with uncertainty) to produce control-relevant bladder states (e.g., volume trajectory, pressure trend, detrusor-overactivity risk with confidence). These estimates drive a controller/optimizer that issues stimulation commands, while a safety supervisor continuously enforces hard limits and can override or restore a clinician-defined safe mode when signal quality degrades, confidence is low, or failures are detected.

Conceptually, multiple control policies can operate on the same estimated states but adjust different stimulation variables. Threshold-triggered control initiates stimulation when a state crosses a preset boundary (e.g., high detrusor-overactivity risk or rapid pressure rise), primarily selecting timing (trigger onset) and duration (burst length), with fixed waveform/intensity if needed. Model predictive control uses short-horizon predictions to choose stimulation under constraints, typically optimizing timing, intensity, and duration over time and, when supported, waveform/pattern. Adaptive dosing updates stimulation based on observed response and tolerance, adjusting intensity and/or waveform parameters (e.g., pulse width/frequency) while maintaining safety limits and refractory periods.

Prospective Validation, Benchmarking, and Translation Readiness

Finally, clinical translation requires rigorous prospective valida-

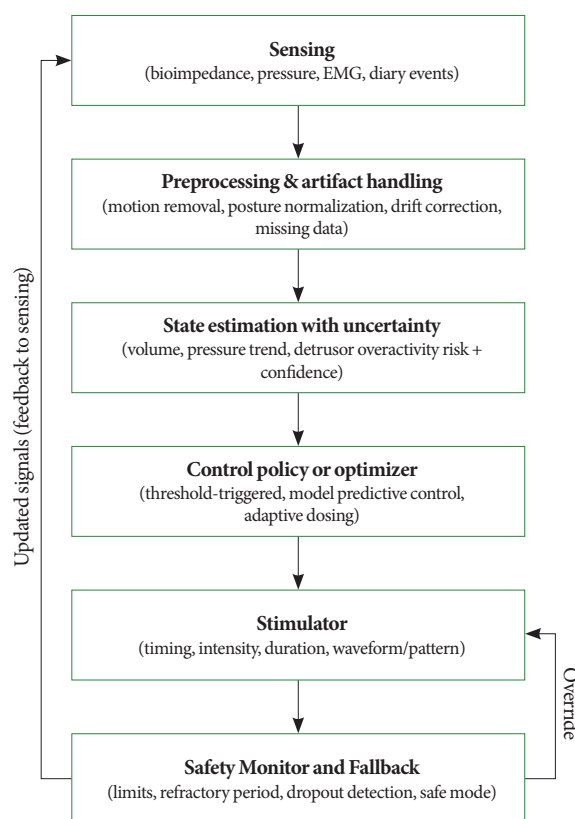


Fig. 5. Closed-loop control architecture for bladder digital-twin-guided neuromodulation. Sensing feeds preprocessing/artifact handling, uncertainty-aware state estimation, and a control policy/optimizer that selects stimulation commands (timing, intensity, duration, waveform/pattern). A safety monitor supervises the loop and can override the stimulator and trigger fallback mode under low-confidence estimates, dropout, or detected failure. EMG, electromyography.

tion, standardized benchmarking, and regulatory readiness [1-3, 12]. Most bladder digital-twin platforms remain at feasibility or early validation stages; wider adoption will require longitudinal human studies under real-world conditions with specified safety and effectiveness endpoints, aligned with how other organ digital-twin programs have moved toward clinical use [1, 3, 16]. Shared evaluation metrics, interoperable data schemas spanning sensing, imaging, and urodynamics, and reproducible benchmarking protocols will be essential for scalability and cross-site comparison [1, 3]. In parallel, privacy-preserving data governance and transparent model versioning should be established early to support deployment and alignment with software-based medical device expectations [1, 3, 12].

CONCLUSION

Bladder digital twins can advance precision neurourology by combining patient-specific modeling with multimodal sensing to estimate bladder state and guide neuromodulation toward closed-loop therapy. Current platforms demonstrate feasibility for mechanics modeling and bladder-state sensing, but translation is limited by ambulatory sensing robustness, long-term drift, real-time state estimation, explicit safety engineering, and prospective human validation. Standardized reporting (accuracy, latency, uptime, drift), hybrid uncertainty-aware estimation, fail-safe control constraints, and longitudinal ambulatory studies with clinical endpoints are therefore needed to move bladder digital twins from prototypes to clinically deployable closed-loop systems.

AUTHOR CONTRIBUTION STATEMENT

- Conceptualization: *THN, EP*
- Data curation: *THN*
- Formal analysis: *THN*
- Funding acquisition: *THN, EP*
- Methodology: *THN, EP*
- Project administration: *THN, EP*
- Visualization: *THN*
- Writing - original draft: *THN*
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